ReJivan (रीजीवन)

Connected care. Earlier action. · A Personal Nurse for Every Family

Hack for Social Cause 2027 (VBYLD 2027 · MoYAS + IIT Bombay)
Submission Deadline: **15 October 2026**
Theme: Healthcare & Wellbeing · Elderly Care & Ageing
Regional Anchor: Andaman & Nicobar Islands (UT)
Owner: **Samrat** · Teacher Trainee

1. EXECUTIVE SUMMARY: WHAT IS REJIVAN?

ReJivan is an intelligent remote healthcare monitoring system engineered to serve as a tireless, 24/7 personal nurse for elderly relatives living at home and for patients in understaffed hospital rooms (**"Virtual Wards"**).

In India today, millions of elderly parents live alone while their adult children work. Concurrently, government and semi-urban hospitals experience severe nurse-to-patient ratios often exceeding 1:30 during night shifts. When an elderly person falls, suffers a silent stroke, experiences sudden hypoxemia, or enters diabetic coma, hours can pass before discovery. ReJivan closes this critical gap through a **Dual-Layer Shield** backed by an **Autonomous 3-Tier Escalation Ladder**.

Layer 1: Continuous Physiological Shield

Ingests clinical vital signs (Heart Rate, SpO2, Blood Pressure, Temperature, Respiratory Rate, Blood Glucose) from consumer & medical wearables. Analyzes readings against the gold-standard **NEWS2** (National Early Warning Score 2) deterioration engine.

Layer 2: Privacy-First Camera Zones

Uses an overhead room camera sentinel to detect catastrophic physical events: sudden falls, long-lie immobility, or unmonitored bed exit. **100% video-private:** no camera video is ever recorded, stored, or streamed to the cloud. Raw camera frames are turned into mathematical joint coordinates in device RAM and erased immediately.

2. THE PROBLEM MATRIX: THE "SILENT GAP" IN CARE

Medical crises among the elderly rarely present with loud calls for help. They happen in terrifying silence:

Clinical Crisis	Why It Is Often Fatal Today	How ReJivan Intervenes Early
The Post-Fall "Long Lie"	Over 30% of elderly adults fall annually. Inability to reach a phone for 6–12 hours leads to hypothermia, rhabdomyolysis (muscle breakdown), renal failure, and pressure necrosis.	Optical kinematics detect rapid descent velocity and prolonged floor stillness, triggering autonomous alerts within 30 seconds.
Silent Nocturnal Hypoxia	Patients with COPD, pneumonia, or cardiac failure suffer silent oxygen desaturations during deep sleep, progressing to CO2 narcosis and arrest without awakening.	Continuous SpO2 tracking identifies drops below 90% immediately, sounding local awakenings and alerting family.
Understaffed Hospital Wards	Night-shift ward nurses cannot check vitals continuously for 40 beds. Post-operative deterioration between 4-hour manual rounds goes unnoticed.	The Virtual Ward central matrix ranks all beds by real-time NEWS2 score, highlighting deteriorating patients instantly at the nurse station.
Island Healthcare Isolation	In Andaman & Nicobar, outer islands (e.g., Little Andaman) are hours away from tertiary care at GB Pant Hospital in Port Blair. Delays in recognizing crises cause preventable deaths.	Local edge intelligence and 100% offline mobile operations enable immediate local triage and prompt emergency boat/ambulance dispatch.

3. THE AUTONOMOUS 3-TIER EMERGENCY LADDER

Unlike conventional smartwatches that merely vibrate on the user's wrist when they are unconscious, ReJivan executes a deterministic, fail-safe rescue ladder:

Tier 1 (0 to 15s): Primary Family Caregiver — IN-APP ALERT + VOICE CALL

Pushes high-priority alert and auto-initiates phone call to the primary contact (e.g., daughter Priya).

Tier 2 (15 to 45s): Secondary Family / Neighbor Backup AUTOMATED ESCALATION

If primary caregiver is unreachable or doesn't acknowledge within 30s, the system escalates to the backup contact (e.g., Dr. Rajesh or trusted neighbor).

Tier 3 (45s+): Autonomous Emergency Dispatch (108 / 112) LIFE-SAVING DISPATCH

If both family tiers fail, autonomous emergency protocols dispatch government EMS (108 Ambulance / Port Blair Marine Ambulance) transmitting GPS coordinates, medical history, and vital crisis telemetry.

4. DUAL DEPLOYMENT: HOME CARE VS. HOSPITAL "VIRTUAL WARD"

Home Care Mode

FAMILY ACCESS

Tailored for adult children monitoring parents living in Port Blair or outer islands.

- **5 Indian Languages:** English, Hindi (हिन्दी), Bengali (বাংলা), Tamil (தமிழ்), Telugu (తెలుగు).
- **Medication MAR:** Clear adherence tracker with daily dosages and missed-pill alerts.
- **Privacy Radar:** Displays posture status (Standing, Seated, Supine) without exposing video.
- **One-Tap Check-in:** Direct voice reassurance button.

Hospital Virtual Ward

CLINICIAN ACCESS

Dedicated to hospital nurses (e.g., GB Pant Hospital Ward Nurse Shift A).

- **Multi-Bed Telemetry:** Live dashboard tracking Bed 101 to 104 with dynamic acuity rank.
- **NEWS2 Scoring:** Standard Royal College of Physicians clinical scoring.
- **4-Stage Follow-up Lifecycle:** `New` → `Acknowledged` → `Follow-up Recorded` → `Resolved`.
- **Clinical Audit Trail:** Documented nurse action notes compliant with ABDM standards.

5. ARCHITECTURAL COMPONENTS INSIDE THE REJIVAN WORKSPACE

Subsystem	Location in Codebase	Key Capabilities
Judge & Clinician Web Platform	`prototype/` Node.js, Express, React	Serves live dashboard at `localhost:8080` and `rejivan.vercel.app`. Multi-patient isolation (Anita Sharma, Ram Prakash, Ward Nurse). Zero-refresh telemetry polling.
Administrator Simulator	`prototype/public/admin-simulator.html` Route: `/admin/simulator`	Protected by a private, server-verified team developer key. Injects deterministic physiological scenarios (Normal, High HR, Low SpO2, BP Crisis, Recovery). Powers seamless live demonstrations.
Edge Computer Vision Sentinel	`tools/yolo_edge_sentinel.py` Python, Ultralytics YOLO-Pose	Extracts 17 COCO body keypoints on local GPU (RTX 4060 / GTX 1650). Computes descent velocity, torso tilt, floor immobility, and patient room disappearance.
Offline Native Android App	`app-android/` Kotlin, Jetpack Compose	100% offline-first. Contains embedded VitalSimulator, RulesEngine, MedStore, and Emergency Caller. Built APK ready at `ReJivan_v1.0.apk`.

6. PROTOTYPE HONESTY: WHAT IS REAL VS. SIMULATED

In compliance with Hack for Social Cause 2027 rules and academic ethics, we explicitly disclose the implementation boundary:

✓ FULLY REAL & OPERATIONAL SOFTWARE

- Full web applications, dashboards, routing, and role authentication.
- Deterministic clinical rules engine and NEWS2 acuity calculation.
- 4-stage clinical alert follow-up lifecycle and persistent database notes.
- YOLO-Pose AI computer vision pipeline running on local graphics hardware.
- Native offline Android application with embedded rules engine.

⚠ SIMULATED FOR SAFE LIVE DEMONSTRATION

- **Hardware Sensor BLE Pairing:** Vitals are injected via the Admin Simulator so judges can evaluate emergency alarms without wearing medical hardware.
- **Carrier Telecom Dialing:** Escalation logic, timing, retries, and dispatch sirens are 100% real, but actual cellular tower connections to government 108/112 lines are simulated to prevent false emergency dispatches.

- Real-time dual-interface state synchronization (Admin → Judge).

7. HOW TO DEMONSTRATE REJIVAN TO EVALUATORS (STEP-BY-STEP)

1. **Step 1: Open Judge View** — Open `http://localhost:8080` (or `https://rejivan.vercel.app`). Log in as Hospital Nurse (`wardnurse@demo.in` / `demo123`). Point out the clean, professional ward telemetry grid with Bed 101–104.
2. **Step 2: Open Admin Simulator** — In a separate browser tab or phone, open `/admin/simulator` and enter the private developer key shared with the presenting team.
3. **Step 3: Trigger Physiological Crisis** — Click **"Elevated HR"** (138 bpm) or **"Low SpO2"** (88%).
4. **Step 4: Observe Live Sync** — Without refreshing the Judge window, observe Bed 101 turn amber/red within 1.8 seconds. A critical alert is prepended displaying the exact clinical condition: "*Heart Rate critical elevation: 138 bpm (threshold: >100 bpm)*".
5. **Step 5: Demonstrate Nurse Workflow** — On the Judge dashboard, click **"Acknowledge"**, type clinician notes in **"Document Follow-up"**, and mark **"Resolved"** to complete the medical audit cycle!